REPIF documentation

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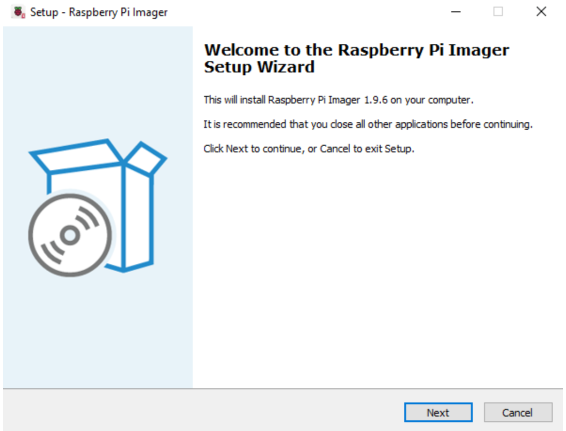
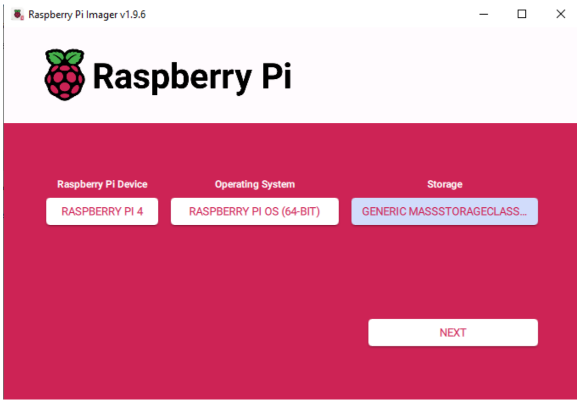
Contents

Raspberry PI

Firstly, I went on the official raspberry Pi website and downloaded the Raspberry Pi Imager.

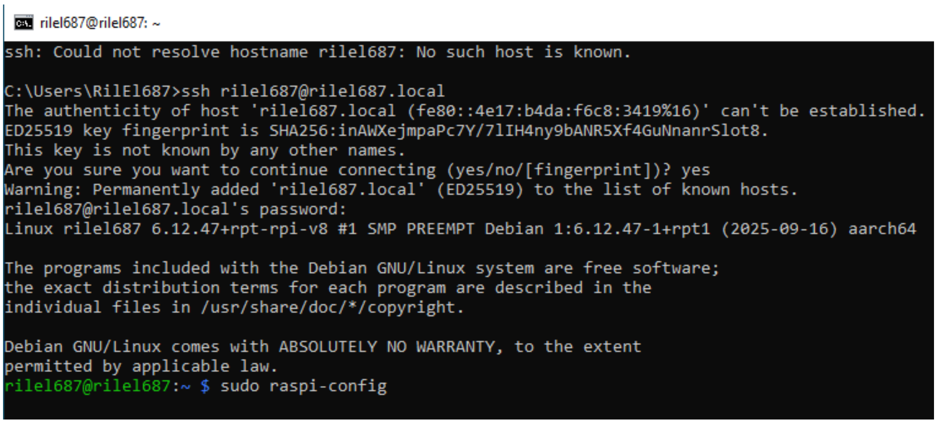


Once done I setup the correct settings for the download and finished the installation.



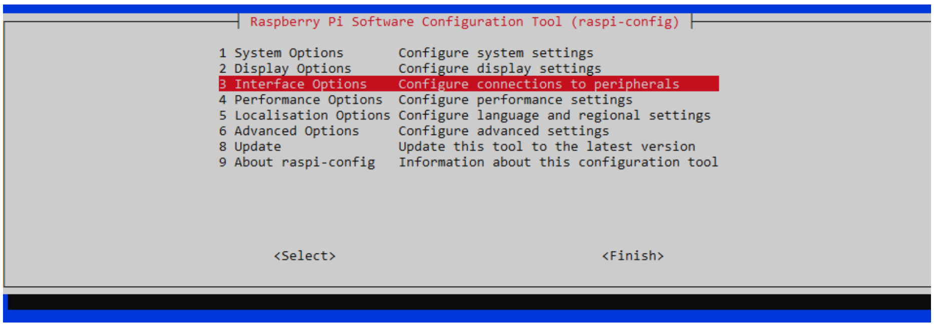
After the installation was complete, I configured the needed settings to implement in the micro SD card.

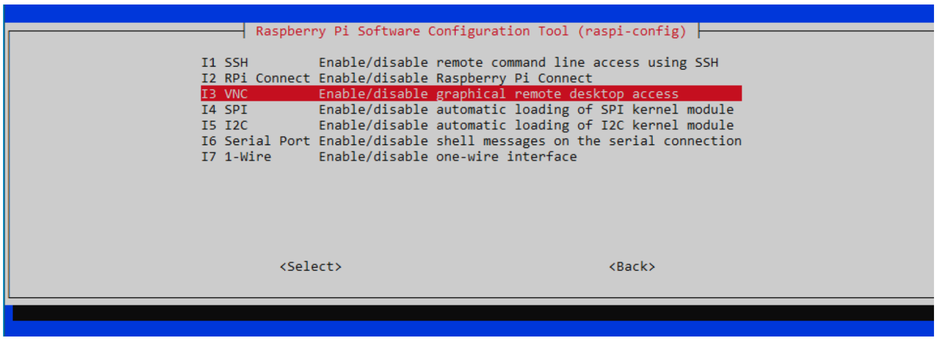
Now the next step was to access it by VNC, so I went into the raspberry configuration like so:



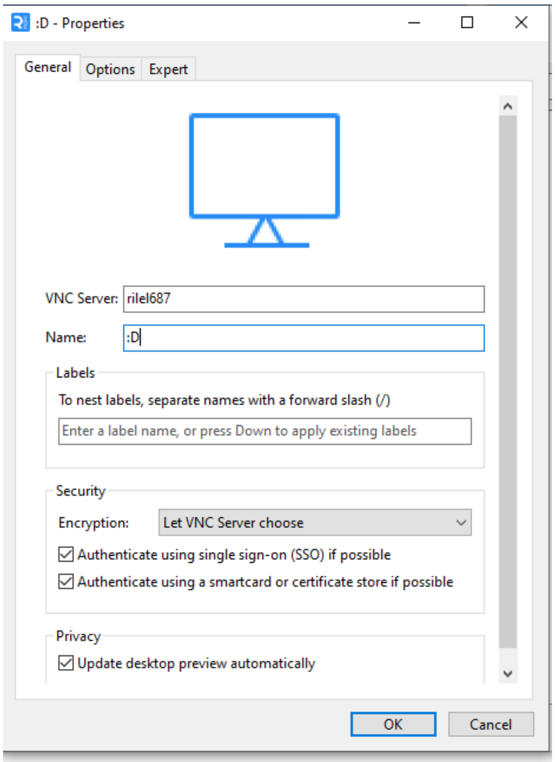

```
ssh: Could not resolve hostname rilel687: No such host is known.

C:\Users\RilEl687>ssh rilel687@rilel687.local
The authenticity of host 'rilel687.local (fe80::4e17:b4da:f6c8:3419%16)' can't be established.
ED25519 key fingerprint is SHA256:inAWXejmpaPc7Y/7lIH4ny9bANR5Xf4GuNnanrSlot8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'rilel687.local' (ED25519) to the list of known hosts.
rilel687@rilel687.local's password:
Linux rilel687 6.12.47+rpt-rpi-v8 #1 SMP PREEMPT Debian 1:6.12.47-1+rpt1 (2025-09-16) aarch64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
rilel687@rilel687:~ $ sudo raspi-config
```

Once inside I selected Interface Options, then VNC.

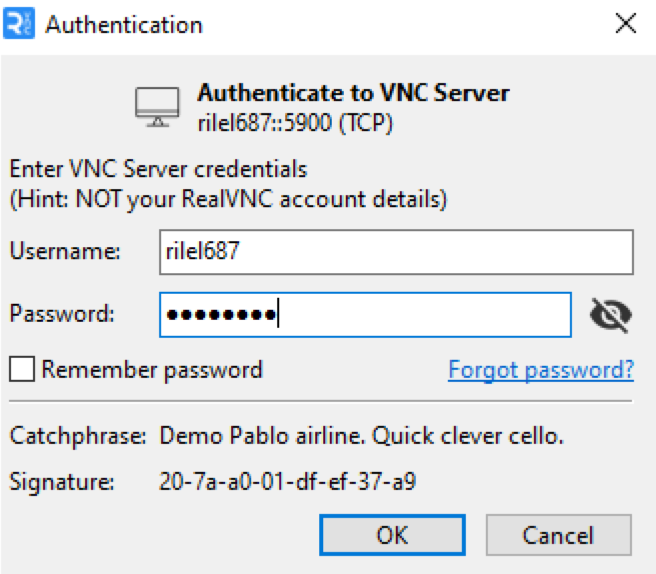




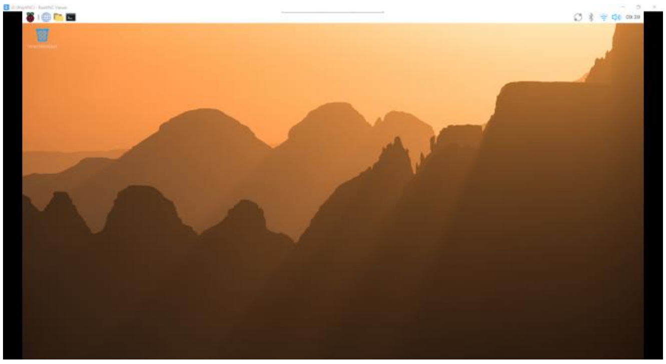
Now onto my VNC I selected file, new connection and I added the used username with the Name of my choice



Then I needed to put in the username and password that I set up earlier.

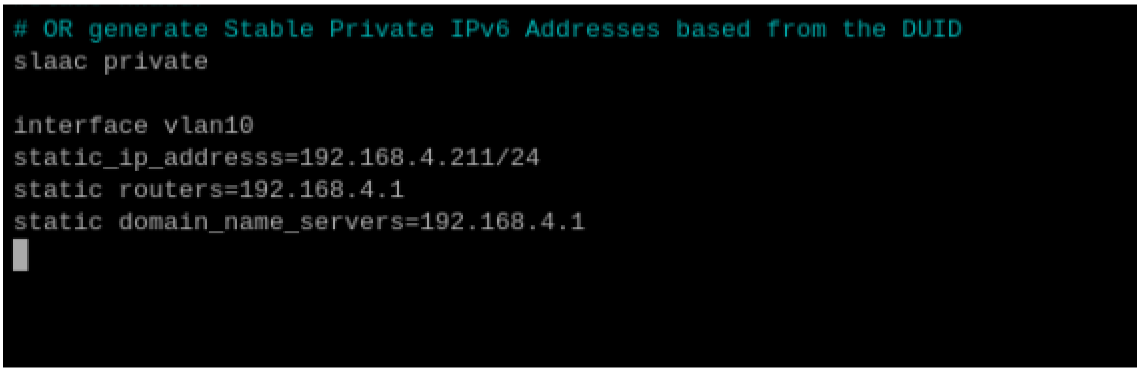


If the password and username match it should let you enter in a few seconds.



Now to add a static IP address I typed the command sudo nano /etc/dhcpcd.conf and accessed the file.

Once inside at the bottom of it I added the following lines:



Once done to enable the i2C interface I entered the configuration with “sudo raspi-config”.

Then I selected Interface Options, I2C and enabled it.

```
Raspberry Pi Software Configuration Tool (raspi-config)

1 System Options       Configure system settings
2 Display Options      Configure display settings
3 Interface Options    Configure connections to peripherals
4 Performance Options  Configure performance settings
5 Localisation Options Configure language and regional settings
6 Advanced Options     Configure advanced settings
8 Update               Update this tool to the latest version
9 About raspi-config   Information about this configuration tool
```

```
Raspberry Pi Software Configuration Tool (raspi-config)

I1 SSH         Enable/disable remote command line access using SSH
I2 RPi Connect Enable/disable Raspberry Pi Connect
I3 VNC         Enable/disable graphical remote desktop access
I4 SPI         Enable/disable automatic loading of SPI kernel module
I5 I2C         Enable/disable automatic loading of I2C kernel module
I6 Serial Port Enable/disable shell messages on the serial connection
I7 1-Wire      Enable/disable one-wire interface
```

Docker

Now to install Docker, I typed the following commands and completed the installation:

```
rilel687@rilel687:~ $ curl -fsSL https://get.docker.com -o install-docker.sh
rilel687@rilel687:~ $ cat install-docker.sh
#!/bin/sh
set -e
# Docker Engine for Linux installation script.
```

```
rilel687@rilel687:~ $ sh install-docker.sh --dry-run
# Executing docker install script, commit: 7d96bd3c5235ab2121bcb855dd7b3f3f37128ed4
apt-get -qq update >/dev/null
DEBIAN_FRONTEND=noninteractive apt-get -y -qq install ca-certificates curl >/dev/null
install -m 0755 -d /etc/apt/keyrings
curl -fsSL "https://download.docker.com/linux/debian/gpg" -o /etc/apt/keyrings/docker.asc
chmod a+r /etc/apt/keyrings/docker.asc
echo "deb [arch=arm64 signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/debian trixie stable" > /etc/apt/
apt-get -qq update >/dev/null
DEBIAN_FRONTEND=noninteractive apt-get -y -qq install docker-ce docker-ce-cli containerd.io docker-compose-plugin docker-ce-rootl
rilel687@rilel687:~ $ sudo sh install-docker.sh
# Executing docker install script, commit: 7d96bd3c5235ab2121bcb855dd7b3f3f37128ed4
+ sh -c apt-get -qq update >/dev/null
+ sh -c DEBIAN_FRONTEND=noninteractive apt-get -y -qq install ca-certificates curl >/dev/null
+ sh -c install -m 0755 -d /etc/apt/keyrings
+ sh -c curl -fsSL "https://download.docker.com/linux/debian/gpg" -o /etc/apt/keyrings/docker.asc
+ sh -c chmod a+r /etc/apt/keyrings/docker.asc
```

To check if it was correctly installed and running, I typed “sudo systemctl status docker”.

```
rilel687@rilel687:~ $ sudo systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
     Active: active (running) since Thu 2025-11-13 10:45:30 CET; 2min 58s ago
```

Now I created the folders /opt/station and /var/station with the correct permissions:

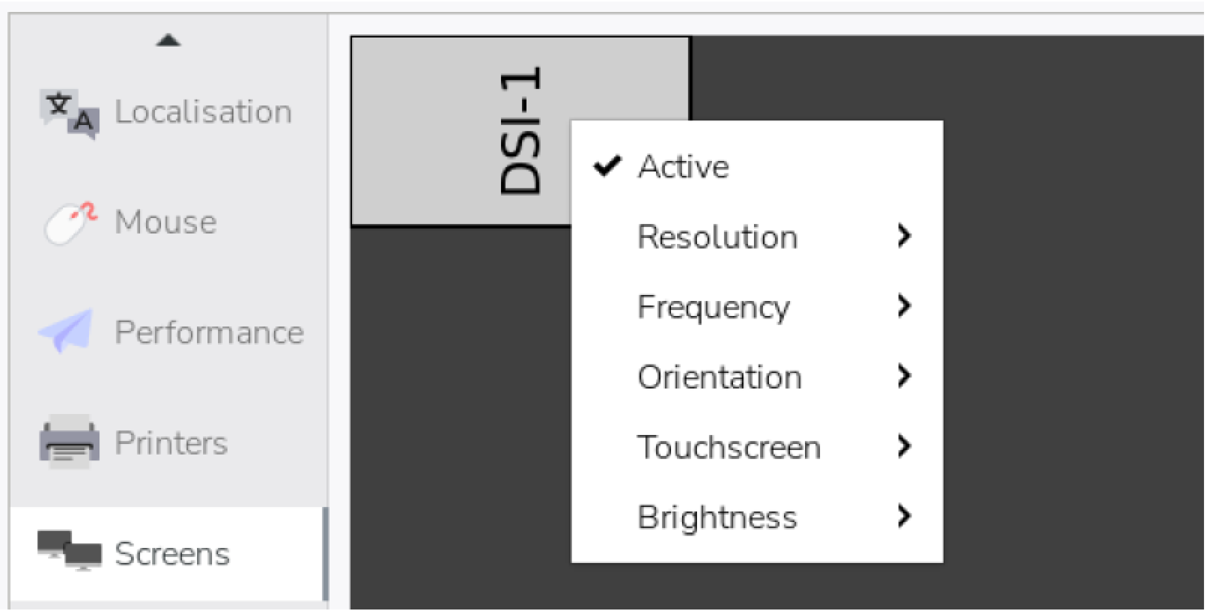
```
rilel687@rilel687:~ $ sudo mkdir -p /opt/station
rilel687@rilel687:~ $ sudo mkdir -p /var/station
```

```
rilel687@rilel687:~ $ sudo chmod 755 /opt/station/
rilel687@rilel687:~ $ sudo chmod 755 /var/station
```

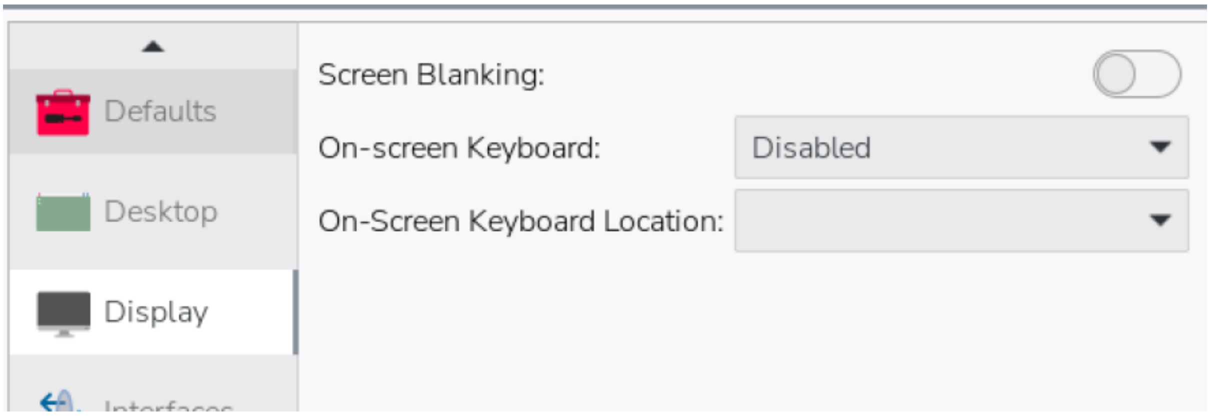
LCD screen connecting

I plugged in the cables form the raspberry pi to my lcd screen and screwed them together.

Now after booting it up I changed the display to landscape by going to the control center , screens, right clicked on my display and changed the orientation to Right.



Then I also disabled the on screen keyboard by going to the display and disabling it.



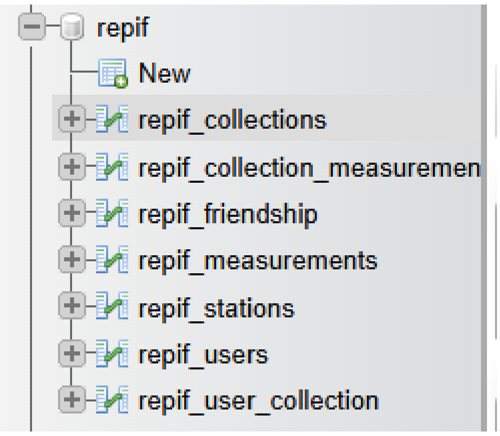
Copy and Move Resources

I downloaded the resources from teams in my raspberry pi, then unzipped the folder, and entered the following commands to put them in their respective folders.

```
ilel687@rilel687:~/Downloads/Ressourcen $ sudo mv Firmware\ und\ Scripts/ /opt/station/
```

```
rilel687@rilel687:~/Downloads/Ressourcen $ sudo mv Station-Webseite/ /var/www/html/
```

REPIF Database



←T→	collection_id	collection_name	collection_desc
Edit Copy Delete	10	Office Sensors	Sensors located in the main office
Edit Copy Delete	11	Outdoor Sensors	Sensors placed outside
Edit Copy Delete	12	Basement Experiments	Experimental sensor logs

INSERT INTO repif_users (user_id, username, userfullname, useremail, user_role) VALUES

(1, 'alice', 'Alice Johnson', 'alice@example.com', 'USER'),

(2, 'bob', 'Bob Smith', 'bob@example.com', 'USER'),

(3, 'admin', 'Admin User', 'admin@example.com', 'ADMIN');

INSERT INTO repif_collections (collection_id, collection_name, collection_desc) VALUES

(10, 'Office Sensors', 'Sensors located in the main office'),

(11, 'Outdoor Sensors', 'Sensors placed outside'),

(12, 'Basement Experiments', 'Experimental sensor logs');

INSERT INTO repif_stations (serial_id, serial_number, station_name, station_desc, user_id) VALUES

(100, 'SN-A1', 'Office Station A', 'Main office environment station', 1),

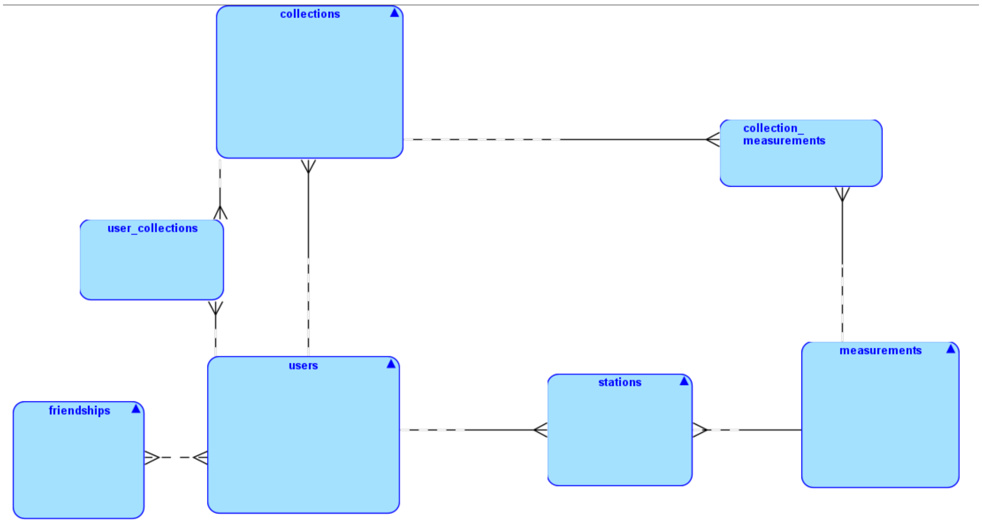
(101, 'SN-B2', 'Outdoor Station B', 'Weather monitoring station', 1),

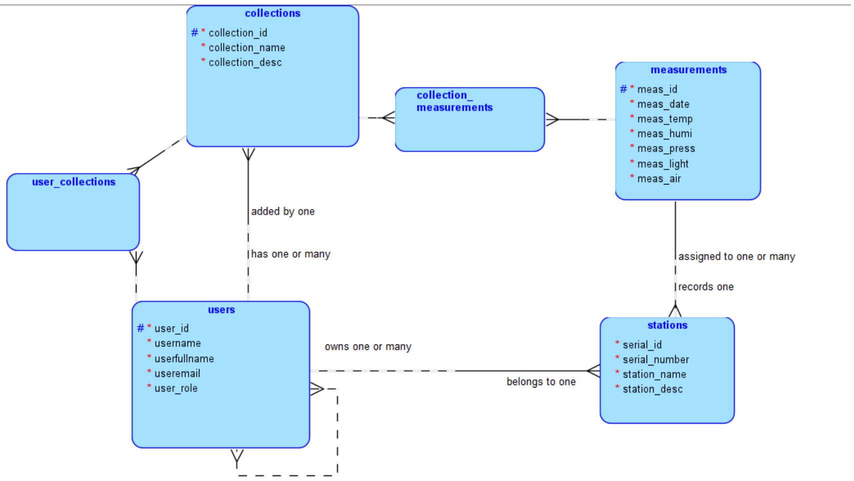
(102, 'SN-C3', 'Basement Station C', 'Experiment station', 2);

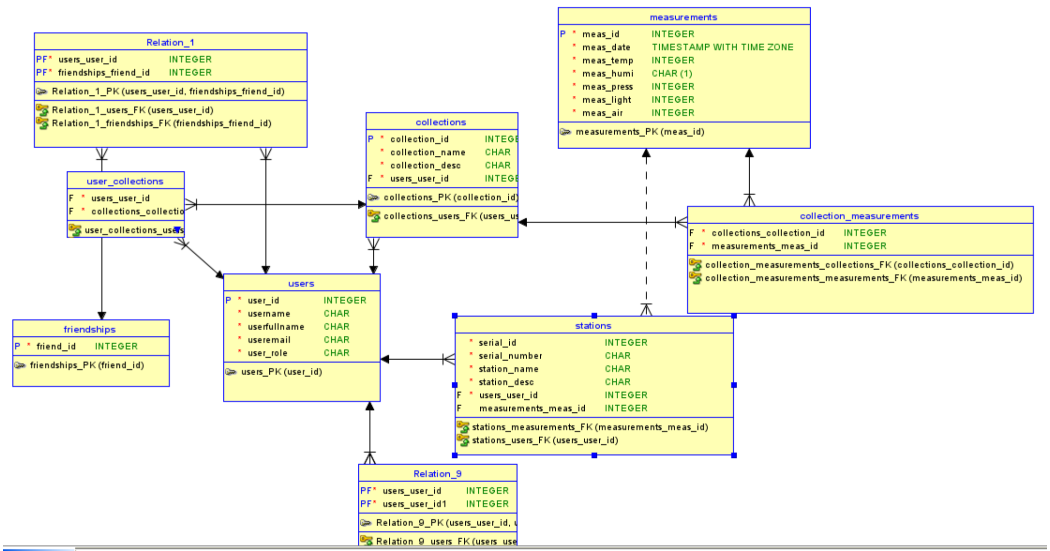
INSERT INTO repif_measurements (meas_id, serial_id, meas_date, meas_temp, meas_humi, meas_press, meas_light, meas_air) VALUES

```
(1000, 100, NOW(), 22.5, 45.2, 1012.3, 300.0, 12.1),
(1001, 100, NOW(), 23.1, 44.8, 1011.9, 305.0, 12.0),
(1002, 101, NOW(), 18.4, 60.3, 1008.1, 500.0, 20.2),
(1003, 102, NOW(), 20.0, 55.0, 1015.0, 100.0, 15.0);

INSERT INTO repif_user_collection (collection_id, user_id) VALUES
(10, 1),
(11, 1),
(12, 2);

INSERT INTO repif_collection_measurements (collection_id, meas_id, user_id) VALUES
(10, 1000, 1),
(10, 1001, 1),
(11, 1002, 1),
(12, 1003, 2);

INSERT INTO repif_friendship (friend_id) VALUES
('1-2'),
('1-3'),
('2-3');
```

CDM LDM

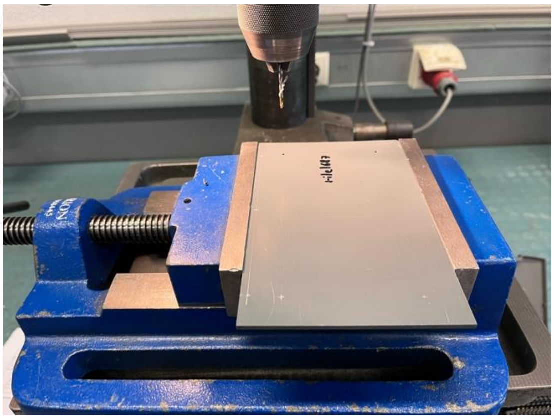

collections
collection_
measurements
user_collections
users
stations
measurements
friendships


collections
* collection_id
* collection_name
* collection_desc
collection_
measurements
measurements
* meas_id
* meas_date
* meas_temp
* meas_humi
* meas_press
* meas_light
* meas_air
user_collections
added by one
has one or many
assigned to one or many
records one
users
* user_id
* username
* userfullname
* useremail
* user_role
owns one or many
belongs to one
stations
* serial_id
* serial_number
* station_name
* station_desc


Relation_1
PF* users_user_id INTEGER
PF* friendships_friend_id INTEGER
Relation_1_PK (users_user_id, friendships_friend_id)
Relation_1_users_FK (users_user_id)
Relation_1_friendships_FK (friendships_friend_id)
measurements
P * meas_id INTEGER
* meas_date TIMESTAMP WITH TIME ZONE
* meas_temp INTEGER
* meas_humi CHAR (1)
* meas_press INTEGER
* meas_light INTEGER
* meas_air INTEGER
measurements_PK (meas_id)
collections
P * collection_id INTEGE
* collection_name CHAR
* collection_desc CHAR
F * users_user_id INTEGE
collections_PK (collection_id)
collections_users_FK (users_us
user_collections
F * users_user_id
F * collections_collectio
user_collections_users
collection_measurements
F * collections_collection_id INTEGER
F * measurements_meas_id INTEGER
collection_measurements_collections_FK (collections_collection_id)
collection_measurements_measurements_FK (measurements_meas_id)
friendships
P * friend_id INTEGER
friendships_PK (friend_id)
users
P * user_id INTEGER
* username CHAR
* userfullname CHAR
* useremail CHAR
* user_role CHAR
users_PK (user_id)
stations
* serial_id INTEGER
* serial_number CHAR
* station_name CHAR
* station_desc CHAR
F * users_user_id INTEGER
F measurements_meas_id INTEGER
stations_measurements_FK (measurements_meas_id)
stations_users_FK (users_user_id)
Relation_9
PF* users_user_id INTEGER
PF* users_user_id1 INTEGER
Relation_9_PK (users_user_id, u
Relation_9_users_FK (users_use

Workshop - Plastic Base Plate

The first step after getting my plates, was scribing the said lines to make the drilled holes with their intersections. Once i had scribed my plates, i punctured the holes and began the drilling with the correct size and speed.



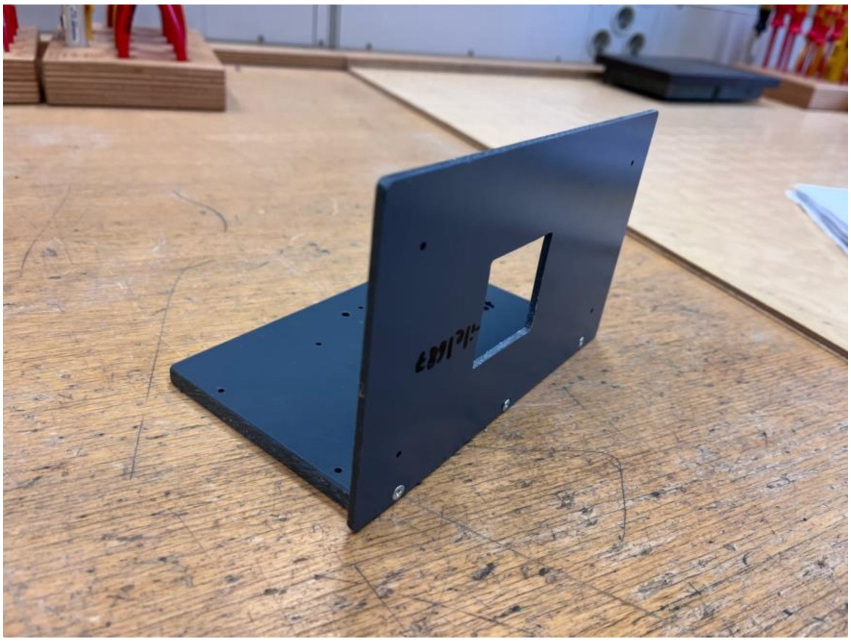
For the square in the middle, i have drilled 4 holes on each corner, once those were finished, i used a small round file, and filed down until a small saw fitted into the intersection. Once that stage reached, i was able to very easly saw down until the next hole,

This step was repeated until the square opened fully.

Once done, i took a bigger file, and made the edges straight and softer to the touch.

Now the final step was to debour the drilled holes so that the screws would later fit smoothly.

Once done I was able to screw the two plates together with the correct screws and it looked like so:

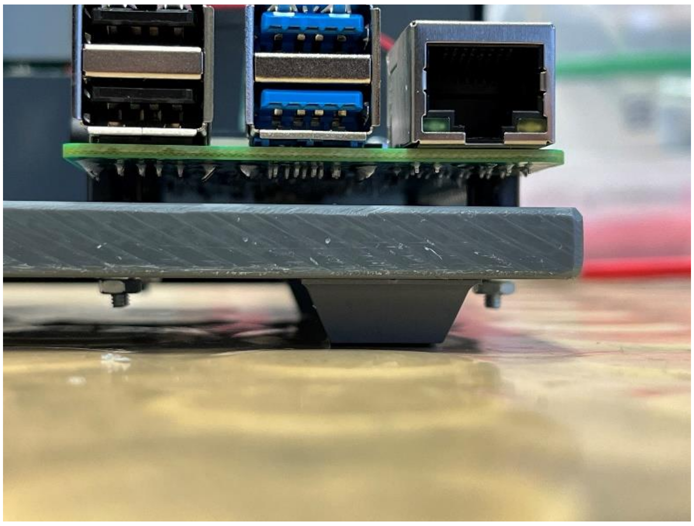


Raspberri Pi

Now i needed to screw the raspberry pi to the plastic base, however I encountered a problem, one of the four holes was about 1-2 milimeters off and I couldn't fit the rapsberri pi with the screws.

So I went back to the drilling machine, made a secondary merged hole almost at the same spot, and was finally able to fit the raspberry pi.

To fix it I needed black separators that avoid friction with the base, a nut, and a thin piece of metal between the base and the nut.



Once screwed tightly, I added little plastic feet to the bottom of the base, to avoid collisions with the lcd screen:

Now I added the lcd screen at the back of the plate, the cables were made to fit in the previously made square.

Once again, I used black plastic separators with screws, and this time the holes were placed perfectly. Once screwed I connected again the cables to the raspberry pi, and this was the final result:

Workshop – Circuit board

List of materials

Name	Quantity
Cable - *(Ground, Live and Neutral)*	1
Cutting plier	1
Soldering station	1
Solder	As much as needed
Wire stripper	1
Cable jacket stripper	1
Heat shrink tubing	4
Heat Gun	1

Steps

Preparation

Firstly, we can cut our **cable** the length we require with our **cutting plier,** making two halves.

Then, we need to take our cable and remove the outer jacketing with a **cable jacket stripper** to expose the shielding.

Figure 1 - Cutting plier with cables cut in half

We need to avoid **damaging the armor** that protects the next layer for later, so gentle pressure against the jacketing and a turn around the surface is enough to detach the part.

Figure 2 - Cable with jacketing removed

Our next step would be to pull back the armor and use once again our **cable jacket stripper** *(if possible a smaller one for easier removal)* to remove the filler that is between our shield and 3 wires.

After this step our 3 wires should be exposed and have **a certain length** for easier access and manipulation.

Figure 3 - Exposed Wires without the filler

Once done we can now even their length out with **cutting pliers** *(make them reach the same length for straight connection)* and remove the primary insulation of our wires to expose the conductor *(copper part)* by using our **wire stripper** for each of them.

Figure 4 - Stripped Wires / Exposed conductor

Soldering

Now onto the soldering part, we can turn up our **soldering station** up to the maximum heat.

While its heating we can line up the wires with each other and verify they have the appropriate length, and we can start applying some **solder** to our conductor part of the wires.

Figure 5 - Soldering station

Before actually soldering the wires with each other we need to make sure that our **heat shrink tubing** is around one of the two wires.

Once done we can now solder all the cables with their respective double and later on, use a **heat gun** to warm up the **heat shrink tubes** around the soldered part to make it tighten.

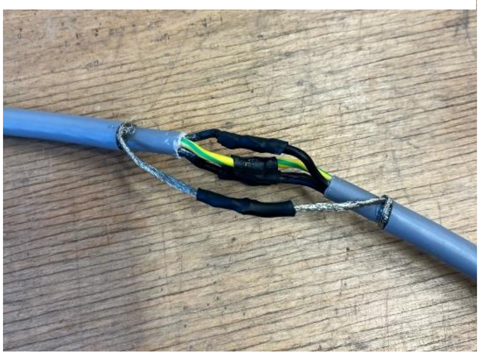


Figure 6 - Final result

For the loose shielding we can twist them and do the exact same process as with our wires.

Issues encountered

Cutting depth

I encountered a few issues in the process, the first one being cutting too deep into the jacketing and damaging the shielding.

Fix - I cut the damaged part and did again that part but delicately this time.

Wire length

My second issue was that I did not leave enough length for my wires to hang out, which lead to more problems such as hardly being able to strip the wires correctly due to the tool's size, and not being able to easly fit my heat shrink tubings into the wires without it disturbing my soldering. With the heat coming from my soldering needle, it began to shrink the tubings and tighten where it should not have.

Fix - I had to make the tubings the exact required length to avoid at most contact with the heat.

Workshop – Powersupply

List of materials

Name	Quantity
Cable - *(Ground, Live and Neutral)*	1
Cutting plier	1
Soldering station	1
Solder	As much as needed
Wire stripper	1
Cable jacket stripper	1
Heat shrink tubing	4
Heat Gun	1

Steps

Preparation

Firstly, we can cut our **cable** the length we require with our **cutting plier,** making two halves.

Then, we need to take our cable and remove the outer jacketing with a **cable jacket stripper** to expose the shielding.

Figure 7 - Cutting plier with cables cut in half

We need to avoid **damaging the armor** that protects the next layer for later, so gentle pressure against the jacketing and a turn around the surface is enough to detach the part.

Figure 8 - Cable with jacketing removed

Our next step would be to pull back the armor and use once again our **cable jacket stripper** *(if possible a smaller one for easier removal)* to remove the filler that is between our shield and 3 wires.

After this step our 3 wires should be exposed and have **a certain length** for easier access and manipulation.

Figure 9 - Exposed Wires without the filler

Once done we can now even their length out with **cutting pliers** *(make them reach the same length for straight connection)* and remove the primary insulation of our wires to expose the conductor *(copper part)* by using our **wire stripper** for each of them.

Figure 10 - Stripped Wires / Exposed conductor

Soldering

Now onto the soldering part, we can turn up our **soldering station** up to the maximum heat.

While its heating we can line up the wires with each other and verify they have the appropriate length, and we can start applying some **solder** to our conductor part of the wires.

Figure 11 - Soldering station

Before actually soldering the wires with each other we need to make sure that our **heat shrink tubing** is around one of the two wires.

Once done we can now solder all the cables with their respective double and later on, use a **heat gun** to warm up the **heat shrink tubes** around the soldered part to make it tighten.

For the loose shielding we can twist them and do the exact same process as with our wires.

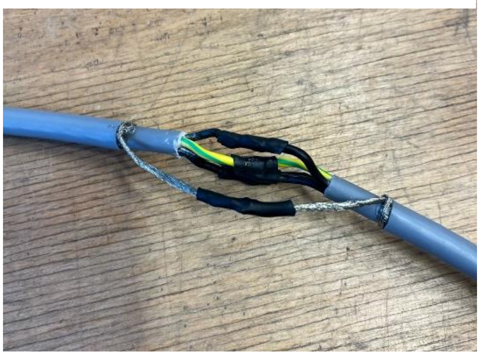


Figure 12 - Final result

Issues encountered

Cutting depth

I encountered a few issues in the process, the first one being cutting too deep into the jacketing and damaging the shielding.

Fix - I cut the damaged part and did again that part but delicately this time.

Wire length

My second issue was that I did not leave enough length for my wires to hang out, which lead to more problems such as hardly being able to strip the wires correctly due to the tool's size, and not being able to easly fit my heat shrink tubings into the wires without it disturbing my soldering. With the heat coming from my soldering needle, it began to shrink the tubings and tighten where it should not have.

Fix - I had to make the tubings the exact required length to avoid at most contact with the heat.

Apache2

My first step was to create a new ubuntu VM and install the required installations.

Firstly, with the command sudo apt -y install apache2 I installed apache2

Then I enabled it:

```
rilel687@rilel687:~$ sudo systemctl enable --now apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
rilel687@rilel687:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-02-26 10:25:05 UTC; 48s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 1600 (apache2)
      Tasks: 55 (limit: 9434)
     Memory: 5.2M (peak: 5.5M)
        CPU: 52ms
     CGroup: /system.slice/apache2.service
             ├─1600 /usr/sbin/apache2 -k start
             ├─1601 /usr/sbin/apache2 -k start
             └─1602 /usr/sbin/apache2 -k start
```

PHP

Now to install all the attachments of php I typed the following long command:

sudo apt -y install php libapache2-mod-php php-mysql php-mbstring php-zip php-gd php-json php-curl php-xml

once installed i restarted apache2

MariaDB

Now onto mariaDB, i installed it with *sudo apt -y install mariadb-server mariadb-client*.

```
MariaDB [(none)]> Create database myapp_db character set utf8mb4 collate utf8mb4_unicode_ci;
Query OK, 1 row affected (0.001 sec)

MariaDB [(none)]> create user 'rilel687'@'localhost' Identified by 'e'
    -> ;
Query OK, 0 rows affected (0.004 sec)

MariaDB [(none)]> grant select, insert, update, delete
    -> on rilel687.*
    -> ^C
MariaDB [(none)]> grant select, insert, update, delete
    -> on myapp_db.*
    -> to 'rilel687'@'localhost';
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]> flus privileges;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use n
s privileges' at line 1
MariaDB [(none)]> flush privileges;
Query OK, 0 rows affected (0.001 sec)

MariaDB [(none)]>
```

I then linked the source of my sql file.

```
Records: 48  Duplicates: 0  Warnings: 0

MariaDB [portableindoorfeedback]> source /var/www/myapp/sf_REPIF1/sql/portableindoorfeedback.sql;
```

And created the test user.

```
MariaDB [portableindoorfeedback]> Create user 'usertest'@'localhost' identified by 'usertest123.';
Query OK, 0 rows affected (0.010 sec)

MariaDB [portableindoorfeedback]> SHOW DATABASES;
+------------------------+
| Database               |
+------------------------+
| information_schema     |
| myapp_db               |
| mysql                  |
| performance_schema     |
| phpmyadmin             |
| portableindoorfeedback |
| sys                    |
+------------------------+
7 rows in set (0.001 sec)

MariaDB [portableindoorfeedback]> grant select, insert, update, delete on portableindoorfeedback.* to 'usertest'@'localhost';
Query OK, 0 rows affected (0.008 sec)

MariaDB [portableindoorfeedback]> flush privileges
    -> ;
Query OK, 0 rows affected (0.003 sec)
```

Sudo nano /etc/apache2/sites-avaiable /repif.conf

```
<VirtualHost *:80>

        ServerName myapp.local
        DocumentRoot /var/www/myapp

        <Directory /var/www/myapp>
                AllowOverride All
                Require all granted
        </Directory>

</VirtualHost>
```

sudo a2ensite myapp.conf
sudo a2dissite 000-default.conf
sudo a2enmod rewrite
sudo systemctl reload apache2

PHPMyAdmin

i installed phpmyadmin with sudo apt install phpmyadmi, and chose apache2, and entered a the root password

192.168.%

me

Fixing measure.py

As my measure.py was breaking after a few seconds, i managed to fix it by adding sleep 20 in my autostart, it later ran smoothly.

```
mkdir -p /tmp/station
python3 /opt/station/Firmware_und_Scripts/measure.py > /tmp/station/result.csv &
sudo systemctl start docker
sleep 3
docker start php
sleep 5

sleep 20
chromium http://localhost:8080 \
--kiosk --noerrdialogs --disable-infobars --no-first-run \
--enable-features=OverlayScrollbar --start-maximized &
```

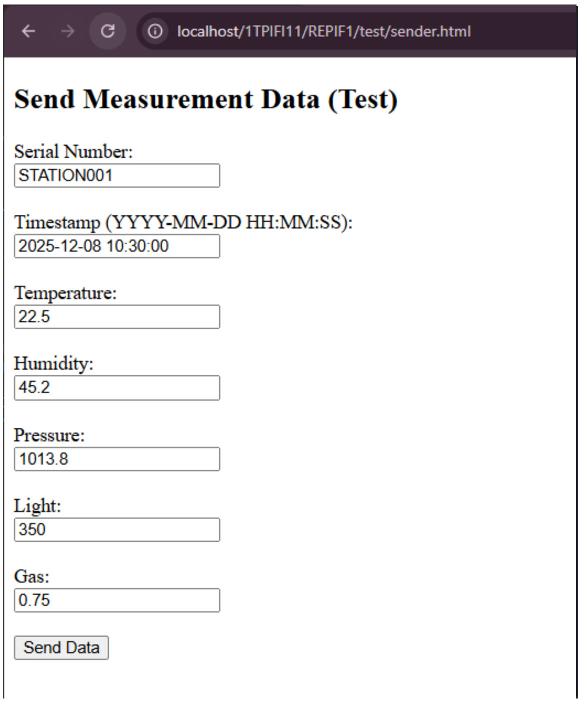
Receiver

I firslty created a script that would, receive the measurements to later on store them into my database more precisely into the table measurements, in the worse case where the values would not be found, it would replace it by 0 and sample data to avoid further problems:

```php
<?php
//connecting to the database
$host = "localhost";
$dbname = "portableindoorfeedback";
$username = "root";
$password = "";
$conn = new mysqli($host, $username, $password, $dbname);
//if it fails :(
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
//making my variables and adding default values if they are not set
$temperature = $_POST['temperature'] ?? 0.0;
$humidity    = $_POST['humidity'] ?? 0.0;
$pressure    = $_POST['pressure'] ?? 0.0;
$light       = $_POST['light'] ?? 0.0;
$gas         = $_POST['gas'] ?? 0.0;
$station     = $_POST['station_serial'] ?? "ST-001";
$timestamp   = $_POST['timestamp'] ?? date("Y-m-d H:i:s");

//inserting into the database
$sqlInsert = $conn->prepare("INSERT INTO measurement
(temperature, humidity, pressure, light, gas, timestamp, fk_station_records) VALUES (?,?,?,?,?,?,?)");
$sqlInsert->bind_param("ddddds", $temperature, $humidity, $pressure, $light, $gas, $timestamp, $station);
$sqlInsert->execute();

?>
```

Sample Sender

I then created a sender.html to replace the measure.py temporarely to see if my receiver.php was able to receive sample data and store it correctly into the database, here is the script and page:




```html
<!DOCTYPE html>
<html>
<head>
    <meta charset="UTF-8">
    <title>Measurement Sender Test</title>
</head>
<body>

<h2>Send Measurement Data (Test)</h2>

<form method="POST" action="receiver.php">

    Serial Number:<br>
    <input type="text" name="station_serial" value="STATION001"><br><br>

    Timestamp (YYYY-MM-DD HH:MM:SS):<br>
    <input type="text" name="timestamp" value="2025-12-08 10:30:00"><br><br>

    Temperature:<br>
    <input type="text" name="temperature" value="22.5"><br><br>

    Humidity:<br>
    <input type="text" name="humidity" value="45.2"><br><br>

    Pressure:<br>
    <input type="text" name="pressure" value="1013.8"><br><br>

    Light:<br>
    <input type="text" name="light" value="350"><br><br>

    Gas:<br>
    <input type="text" name="gas" value="0.75"><br><br>

    <input type="submit" value="Send Data">

</form>

</body>
```

Correcting connections

Once it correctly put the desired data into the database, i went back into my config.toml and put the ip of my host machine with the path to the receiver.php.

```toml
station_serial = "ST-001"           # Serial number of the station

interval = 2                        # Time between measurements

toggle_intensity = 100              # The light level at which the LED should be turned on

[i2c]                               # I2C-Addresses of the sensors
bmp280 = 0x76
bh1750 = 0x23
mq135 = 0x48

[destination]                       # Information about the server
ip = "192.168.4.19"                              # IP-Address or Hostname of the Server
path = "/1TPIFI11/REPIF1/test/receiver.php"                     # Path to the script that accepts the measurements
```

Database Server

In my database VM, I went into the following file to change the bind_address:

```
rilel687@rilel687:/etc/mysql/mariadb.conf.d$ _
```

```
  GNU nano 7.2                                                   50-server.cnf *
#
# These groups are read by MariaDB server.
# Use it for options that only the server (but not clients) should see

# this is read by the standalone daemon and embedded servers
[server]

# this is only for the mysqld standalone daemon
[mysqld]

#
# * Basic Settings
#

#user                    = mysql
pid-file                 = /run/mysqld/mysqld.pid
basedir                  = /usr
#datadir                 = /var/lib/mysql
#tmpdir                  = /tmp

# Broken reverse DNS slows down connections considerably and name resolve is
# safe to skip if there are no "host by domain name" access grants
#skip-name-resolve

# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address             = 0.0.0.0
```

Granting the local network access by changing bind adress to 0.0.0.0

Into my Backup Server

I then went into the file .my.cnf which i used to put my testuser credentials and ip

```
e@rilel687:~$ sudo nano .my.cnf
```

```
  GNU nano 7.2
[client]
user=usertest
password=me
host=192.168.4.32
```

Then i went into my crontab with crontab -e and put the following command that happens once every day at 8am and creates a backup of my database:

```
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 8 * * * mysqldump --column-statistics=0 portableindoorfeedback > /backup/DB_Backup_$(date +\%Y-\%m-\%d_\%H-\%M-\%S).sql
```

After testing with every minute delay, here is the result:

```
No modification made
e@rilel687:/backup$ ls
DB_Backup_2026-04-16_09-42-33.sql  DB_Backup_2026-04-16_09-45-01.sql  DB_Backup_2026-04-16_12-38-01.sql  DB_Backup_2026-04-16_12-43-01.sql
DB_Backup_2026-04-16_09-43-01.sql  DB_Backup_2026-04-16_09-46-01.sql  DB_Backup_2026-04-16_12-39-01.sql  DB_Backup_2026-04-16_12-44-01.sql
DB_Backup_2026-04-16_09-43-50.sql  DB_Backup_2026-04-16_09-47-01.sql  DB_Backup_2026-04-16_12-40-01.sql  DB_Backup_2026-04-16_12-45-01.sql
DB_Backup_2026-04-16_09-44-01.sql  DB_Backup_2026-04-16_12-35-08.sql  DB_Backup_2026-04-16_12-41-01.sql  DB_Backup_2026-04-16_12-46-01.sql
DB_Backup_2026-04-16_09-44-24.sql  DB_Backup_2026-04-16_12-37-01.sql  DB_Backup_2026-04-16_12-42-01.sql  DB_Backup_2026-04-16_12-47-01.sql
e@rilel687:/backup$
```

FTP setup

Firstly, I updated and did the installation:

Sudo alt update, sudo apt install vsftpd -y

Then, I checked if my ftp was already enabled like so:

Sudo systemctl status vsftpd

Next, I went into the following file to correct 4 allowances:

Sudo nano /etc/vsftpd.conf

```
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=NO
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
```

```
#
# You may restrict local users to their home directories.  See the FAQ for
# the possible risks in this before using chroot_local_user or
# chroot_list_enable below.
chroot_local_user=NO_
#
```

Finally, i tested if the ftp was working and took as example my backup files:

```
e@rilel687:/$ ftp 192.168.4.59
Connected to 192.168.4.59.
220 (vsFTPd 3.0.5)
Name (192.168.4.59:e): e
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> cd /backup
250 Directory successfully changed.
ftp> ls
229 Entering Extended Passive Mode (|||36955|)
150 Here comes the directory listing.
-rw-rw-r--    1 1000     1000          812 Apr 16 09:42 DB_Backup_2026-04-16_09-42-33.sql
-rw-rw-r--    1 1000     1000        11896 Apr 16 09:43 DB_Backup_2026-04-16_09-43-01.sql
-rw-rw-r--    1 1000     1000          812 Apr 16 09:43 DB_Backup_2026-04-16_09-43-50.sql
-rw-rw-r--    1 1000     1000        11896 Apr 16 09:44 DB_Backup_2026-04-16_09-44-01.sql
-rw-rw-r--    1 1000     1000          812 Apr 16 09:44 DB_Backup_2026-04-16_09-44-24.sql
-rw-rw-r--    1 1000     1000        11896 Apr 16 09:45 DB_Backup_2026-04-16_09-45-01.sql
-rw-rw-r--    1 1000     1000        11896 Apr 16 09:46 DB_Backup_2026-04-16_09-46-01.sql
-rw-rw-r--    1 1000     1000        11896 Apr 16 09:47 DB_Backup_2026-04-16_09-47-01.sql
-rw-rw-r--    1 1000     1000        11425 Apr 16 12:35 DB_Backup_2026-04-16_12-35-08.sql
-rw-rw-r--    1 1000     1000        11896 Apr 16 12:37 DB_Backup_2026-04-16_12-37-01.sql
```